

UNIT 6

EARTH'S RESOURCES

Learning objectives:

- Define natural resources and give examples (rocks, soil, water, plants).
- Explain uses of rocks, soil, water, and wind in daily life and energy production.
- Describe how plants and animals provide food and materials (cotton, wood, leather).
- Define fuel and explain how burning fuels (wood, coal, oil) produces heat.
- Identify Pakistan's energy sources (water, wind, oil, solar) and natural resources (gas, coal, minerals).
- Recognize key crops and fruits grown in Pakistan (wheat, rice, mangoes).

Natural Resources

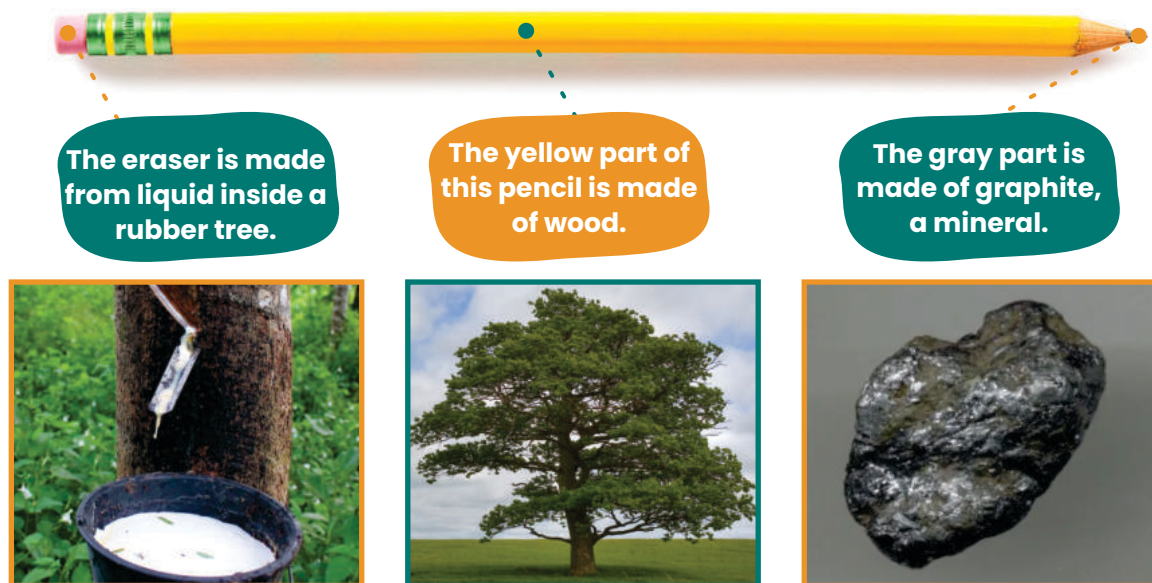
• What Are Natural Resources?

A **natural resource** is something from Earth that people use. Rocks, minerals, plants, soil, and water are natural resources. We use natural resources to make things we use every day.



◀ **Plants, water are natural resources**

Some shirts are made of cotton. Cotton comes from a plant. Your desk might be made of wood. Wood comes from a tree. Look at the diagram below. It shows what we use to make a pencil.



The eraser is made from liquid inside a rubber tree.

The yellow part of this pencil is made of wood.

The gray part is made of graphite, a mineral.

• How Do We Use Rocks and Soil?

Rocks and soil are natural resources. We need rocks and soil to live. Rocks break down and become part of the soil. Plants use the soil to grow. We use plants for food and for making things, such as paper and clothing.



▲ The corn plants need rocks and soil to grow.



We use rocks in other ways. We make concrete by mixing rocks, sand, and water. Concrete is used to make buildings and sidewalks. We use the minerals in sand to make glass.



▲ This building was made with rocks.



▲ The blue parts of this bracelet are turquoise, which is a mineral.

• How Do We Use Water and Wind?

Water and wind are natural resources, too. We use water to drink, cook, and clean. We use it to help plants grow. We also use moving water to make electric power. Electric power lights up and heats our homes.



▲ The water moves over the dam very quickly to generate power.



Have you ever seen a sailboat move across the water or a flag wave in the wind? Then you know that wind can make things move. Wind can also be used to make electric power, just like water.



▲ These windmills are used to make electric power.

Plant and Animal Resources

● How Do We Use Plants?

Many plants are used for food. Fruits, vegetables, and seeds come from plants. The seeds of some plants are called grains. Grains can be ground, or crushed into flour. The flour is then used to make cereal and bread.



▲ First, the seeds, or grains, inside the wheat are taken out.



▲ A mill crushes the grains into flour.



▲ The flour can be used to make bread.



We use plants in many other ways. Some rugs and clothes are made from cotton plants. We use wood from trees to make buildings, furniture, and paper.



Cotton



Wood

• **How Do People Use Animals?**

Animals are natural resources, too. Many people eat chicken, fish, and other animals. The milk from cows, goats, and sheep is used to make butter and cheese. Leather is made from animal skins. Shoes and coats can be made from leather.



Milk



Boot



Beef



Cheese



A **fuel** is something that gives off heat when it burns. We burn wood to give us heat. We also burn coal and oil. Long ago, coal and oil formed underground from dead plants and animals. Now we use coal and oil to cook and heat our homes.



▲ **Gasoline is a fuel made from oil.**
Cars and trucks burn gasoline to move

Resources of Pakistan

How Does Pakistan Get Energy?

People in Pakistan use water, wind, and oil to make electric power. They also use solar power, which is energy from the Sun. Special machines, called **solar panels**, change sunlight into electric power.



▲ **Solar panels capture sunlight to make electricity.**



In Pakistan, there is natural gas and coal under the ground. Workers find it and send it through pipes to buildings. Natural gas can be burned to heat buildings and cook food, while coal is used in power plants to generate electricity.



▲ This person is cooking with natural gas.

What Are Other Natural Resources in Pakistan?

Pakistan has a diverse climate, with sunny plains and fertile valleys. The soil is rich and good for farming. Farmers grow many different crops, such as wheat, rice, cotton, and sugar cane. People in Pakistan and around the world enjoy fruits like mangoes, oranges, and dates from Pakistan.



▲ Mangoes grow well in Pakistan.



Pakistan has many other natural resources. People dig for salt, gypsum, limestone, and copper. The country is famous for its **pink Himalayan salt**, mined from the Khewra Salt Mine. In the past, Pakistan has also been known for its gemstone mining, including emeralds, rubies, and sapphires.



Test Your Understanding

Q1: What is a natural resource? Give three examples.

Ans: _____

Q2: Why do we use natural resources?

Ans: _____

Q3: How do we use rocks and soil in everyday life?

Ans: _____

Q4: Which parts of a pencil come from a tree?

Ans: _____



Q5: What are some ways we use plants in our daily lives?

Ans: _____

Q6: How do we make concrete?

Ans: _____

Q7: What type of power is created when water and wind move?

Ans: _____

Q8: How do people use water and wind to generate power?

Ans: _____



Q9: What is solar power and how is it used?

Ans: _____

Q10: How is natural gas used in homes?

Ans: _____



Fill in the blanks.

1. _____ is something that gives off heat when it burns.
2. _____ power uses sunlight to make electricity.
3. _____ and soil are natural resources that help plants grow.
4. _____ comes from plants and is used to make paper and furniture.
5. _____ is made from animal skins and used to make shoes and coats.
6. Pakistan's soil is rich and good for _____.



Tick (✓) for True and cross (x) for False statements.

1. Solar power comes from the wind. ()

2. Natural gas can be burned to heat buildings ()

and cook food.

3. We use rocks to make glass and concrete. ()

4. Plants are only used for food. ()

5. People in Pakistan and around the world enjoy ()

fruits like mangoes, oranges, and dates from Pakistan.

6. Leather is made from plants. ()

7. Pakistan is famous for its pink Himalayan salt, ()

mined from the Khewra Salt Mine.



Choose the correct answer:

1. Which of the following is a natural resource?

- a) Plastic
- b) Cotton
- c) Glass
- d) Concrete

2. What do we use to make electric power from sunlight?

- a) Wind turbines
- b) Solar panels
- c) Water wheels
- d) Coal

3. Which of the following gives off heat when it burns?

- a) Fuel
- b) Soil
- c) Water
- d) Wind

4. What is used to make buildings, sidewalks, and roads?

- a) Soil
- b) Sand
- c) Concrete
- d) Wood



5. Which of the following crops is commonly grown in Pakistan?

- a) Blueberries
- b) Grapes
- c) Pineapples
- d) Mangoes



Match the Column

Solar power

Using sunlight to make electric power

Fuel

Made from mixing rocks, sand, and water

Natural gas

Something that gives off heat when it burns

Concrete

Made from animal skins

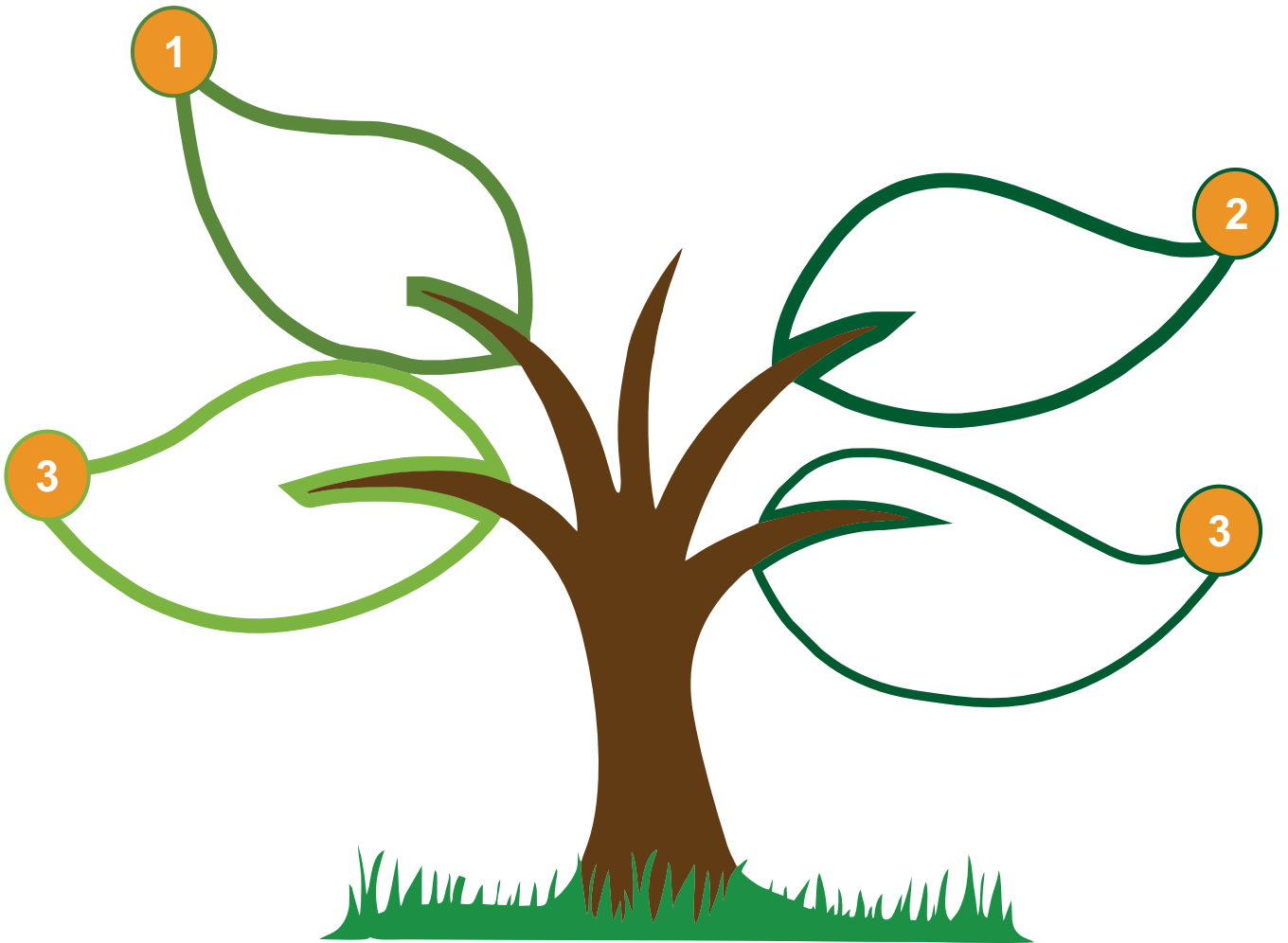
Leather

Used to heat buildings and cook food



HOW ARE PLANTS USED?

Name four ways that plants are used.





Activity 1: Renewable Energy Experiment



ACTIVITY

Create a simple solar oven using a cardboard box, aluminum foil, and plastic wrap. Place a small piece of chocolate or marshmallow inside and observe how the sun's energy melts it. Discuss how solar energy can be used to cook food and generate electricity.



Activity 2: Natural Resource Hunt



ACTIVITY

Go on a scavenger hunt around your home or school to find items made from natural resources (e.g., wooden furniture, cotton clothes, glass windows). List the items and identify the natural resource they come from.





Activity 3: Collage Work



ACTIVITY

Create a collage using pictures of different products made from plants (e.g., food, clothing, furniture). Label each item and explain how plants are important in our daily lives.



Activity 4: Rock Sorting Activity



ACTIVITY

Collect different types of rocks and classify them based on their use (e.g., construction, decoration, tools). Discuss how rocks are important natural resources.

